

RESEARCH ARTICLE

Graphene oxide/polyurethane-based composite solid–solid phase change materials with enhanced energy storage capacity and photothermal performance

Jiawei Wang^{1,2}  | Zihua Wu^{1,2}  | Huaqing Xie^{1,2} | Tingting Wang^{1,2} | Yuanyuan Wang^{1,2} | Yueming Huang^{1,2} | Lan Dong^{1,2}

¹School of Energy and Materials, Shanghai Polytechnic University, Shanghai, China

²Shanghai Engineering Research Center of Advanced Thermal Functional Materials, Shanghai, China

Correspondence

Zihua Wu, School of Energy and Materials, Shanghai Polytechnic University, Shanghai 201209, China.
Email: wuzihua@sspu.edu.cn

Funding information

National Natural Science Foundation of China, Grant/Award Numbers: 51876111, 52176081; Natural Science Foundation of Shanghai, Grant/Award Number: 21ZR1424500; Shanghai Local Capacity Building Program, Grant/Award Numbers: 22010500500, 22010500700; Shanghai Rising-Star Program, Grant/Award Number: 21QA1403300; the “Shu Guang” Project of Shanghai Municipal Education Commission and Shanghai Education Development Foundation, Grant/Award Number: 18SG54

Summary

The preparation of phase change materials (PCMs) with high energy storage, thermal conductivity, and photothermal conversion capability is essential for improving solar energy conversion and storage. In this study, graphene oxide (GO) was incorporated into polyurethane (PU) prepared from 4,4'-diphenylmethane diisocyanate (MDI) and polyethylene glycol (PEG). PU/GO composite PCMs exhibit enhanced thermal conductivity ($0.972 \text{ W m}^{-1} \text{ K}^{-1}$) because the isocyanate in MDI can covalently graft with GO. The composite solid–solid PCMs has outstanding stability and thermal reliability. It can stay stable at 80°C for more than 15 min without leaking and has good thermoregulation regulation performance. The thermal energy storage density and photothermal conversion efficiency of PU/GO increase as the amount of PEG and GO increases. The PU/GO composite with 89 wt% PEG and 1.72 wt% GO has a phase change enthalpy of 150.7 J/g and a high photothermal conversion efficiency (95.3%). The composite PU/GO solid–solid PCMs have great potentials for effective energy storage and solar energy utilization.

KEYWORDS

different soft segment content, energy storage capacity, photothermal conversion performance, solid–solid phase change materials, thermal conductivity

1 | INTRODUCTION

Excessive energy consumption in human production activities has accelerated the emission of greenhouse gases and the depletion of fossil fuels, which may eventually lead to serious ecological problems. Solar energy, the one of the most abundant and cleanest renewable energy sources, attracted extensive attention.^{1,2} Photothermal conversion is an important method for capturing, converting, and storing solar energy,³ and its ability is affected by the thermophysical properties of photon

capture and heat storage materials.^{4,5} Photothermal conversion is an effective way to utilize the solar energy, but it is also severely restricted by intermittent solar radiation.² The latent heat of the phase change materials (PCMs) could collect and store the heat energy which is converted by solar energy and the heat energy conversion of PCMs is easy to operate and has great economic feasibility.

Among many organic PCMs, polyethylene glycol (PEG) was usually considered as one of the most promising solid–liquid organic PCMs because of its biodegradability.

bility, isothermal phase change behavior, suitable phase change temperature range, non-corrosive, non-toxic, good chemical and thermal stabilities.^{6–8} However, PEG should be blended with the support material by simple physical methods, microencapsulated in special-sealed containers and chemically bonded to the skeleton material to form PCMs with better chemical and thermal stabilities due to the liquid leakage during the solid–liquid phase transition.^{9–13} The reaction of PEG and isocyanate to form polyurethane (PU) is very attractive, because PU is a representative of solid–solid PCMs. It has the characteristics of no additional container, good heat storage capacity, high strength, no gas or liquid, and easy processing.^{14,15} Polyurethane solid–solid PCMs are used in a wide range of applications including solar energy conversion, building energy saving, thermo-regulated textiles and aerospace.^{16–18} However, PU PCMs generally have a low thermal conductivity in the range of 0.2 to 0.7 Wm^{−1} K^{−1}, which affects the heat transfer of the PCMs and reduces the energy efficiency.^{19,20} Moreover, conventional polyurethane solid–solid PCMs have poor ability in light absorption and can hardly be used in photothermal conversion devices. A facile strategy to fabricate PCMs with enhanced photothermal performance is to incorporate light-absorbing fillers to the PCMs system.²¹

Metals and metal oxides such as silver,²² titanium black,²³ iron oxide,²⁴ and cobalt oxide²⁵ can be used as fillers because of their effective light-absorbing ability and high thermal conductivity. Zhang et al²⁶ prepared an efficient silver nanoparticles-graphene nanosheets/polyethylene glycol(Ag-GNS/PEG) composite. Ag has high visible light selective absorption and infrared reflection based on the local surface plasmon resonance effect, which can enhance the light absorption capacity of Ag-GNS and reduce thermal radiation. The silver nanoparticles and GNS of Ag-GNS/PEG composites improve the thermal conductivity of the composite due to Ag nanoparticles and GNS-functioned synergistic effect (from 0.212 Wm^{−1} K^{−1} to 0.414 Wm^{−1} K^{−1}), and at the same time, the photothermal conversion efficiency is high (η = 88.7% to 92.0%). Graphene oxide (GO) is a typical layered carbon material with high electrical conductivity, high thermal conductivity, large specific surface area, and high surface activity.^{27,28} GO has high effective light absorption in a wide range to achieve energy conversion and storage.²⁹ Moreover, compared with other fillers, GO can interact with isocyanate (–NCO) in PU because of its –CO, –COOH, and –OH groups on the surface.³⁰ Researchers have found that GO can be combined with solvent-based polyurethane (soft PU).³¹ In case of GO and water-based polyurethane composite PCMs,³² the photothermal properties and thermal stability would be improved after adding GO at

a certain extent. Cai et al³³ first synthesized a linear PU with –NCO group-terminated, mixed with GO solution with good dispersibility and reacted, and finally obtained GO/PU composites. The addition of 4 wt% GO can increase the decomposition temperature approach to 50°C. This is due to the covalent interface formed between GO and PU molecular chains. Zhou et al³⁴ synthesized crosslinked reduced graphene oxide/polyurethane solid–solid phase change material (RGO/PUSSPCM) by a simple solvothermal treatment using PEG served as soft segment and hexamethylene diisocyanate biuret (HDIB) served as crosslinking agent. Different from the direct physical mixing method, the reduction of HDIB and GO realized the physical and chemical crosslinking, so that the samples have good stereotypes, and the phase transition enthalpy is increased by 46.7%. Because GO is reduced in situ, the photothermal conversion efficiency can reach 78.7%. Tang et al³⁵ reported polyethylene glycol/graphene oxide aerogel (PEG/GA) composite PCMs by vacuum impregnation. The PEG/GA composite PCMs have high storage density, shape stability, and more effective photothermal energy conversion. Li et al³⁶ chose GO with photothermal conversion characteristics as the photothermal conversion material and prepared GO/PEG composite PCMs by ultrasonic-assisted physical blending. The results show that the photothermal conversion efficiency of GO/PEG composite PCMs was as high as 87.3%. This is mainly due to the effective light absorption characteristics of GO in a wide wavelength range. But the enthalpy values of GO/PEG composites dropped from 135.81 J/g to 84.27 J/g after adding GO. Because the hydrogen bond interaction between GO and PEG molecules would limit the extension of PEG molecules. This results in incomplete crystallization of PEG and the decrease of latent heat of phase change. Therefore, in order to obtain a composite material with higher photothermal efficiency, more specific gravity light-absorbing fillers are needed, but the energy storage capacity would decrease because the interaction between additive and matrix may affect the enthalpy value of composite material. Bao et al³⁷ showed that the phase change enthalpy of PU synthesized with 4,4'-diphenylmethane diisocyanate (MDI) and PEG increases with the increase of soft segment content. This mainly shows that the content of the soft segment has a certain influence on the enthalpy of phase change in PU. However, only PU with less than 60 wt% PEG content remained in solid state above phase change temperatures and the phase change enthalpy is only 65.95 J/g. Zheng et al³⁸ successfully prepared isocyanated graphene oxide nanosheets/waterborne polyurethane (NCO-GO/WPU). The composites show significant

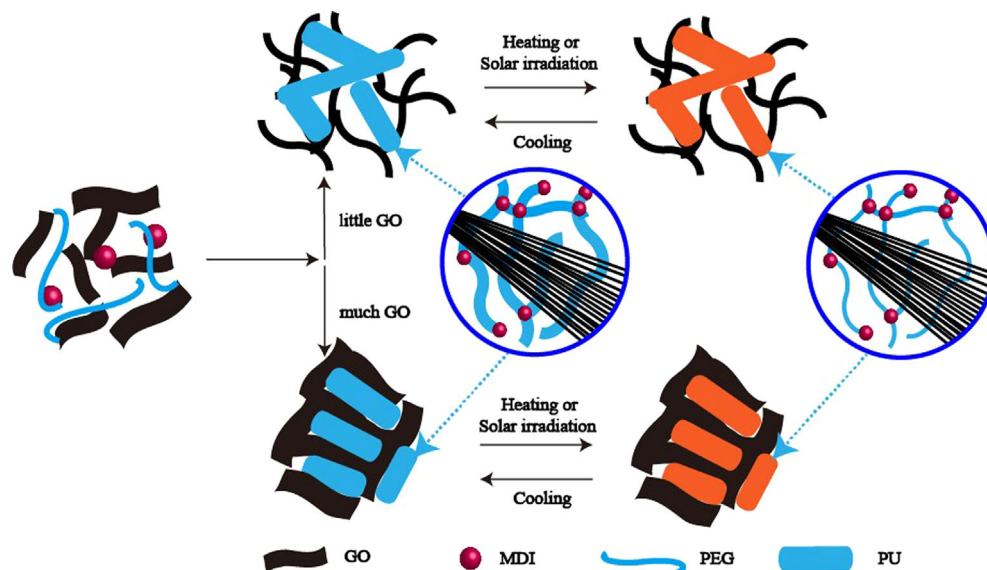


FIGURE 1 Schematic diagram of the mechanism in the PU/GO phase transition process

improvements in performance of tensile strength, initial thermal degradation and hydrophobicity, and its initial degradation temperature increased from 249°C to 288°C. These enhancements in thermal stability were mainly attributed to the two-dimensional structure of NCO-GO and the good interfacial adhesion between NCO-GO and WPU. Kong et al.³⁹ obtained MDI-GO by covalently grafting of GO and MDI, and then added it into PU to prepare MDI-GO/PU composites. The results show that the thermal conductivity of the composites is greatly improved for the interaction of GO and MDI. The thermal conductivity of MDI-GO/PU composites is 51.7% higher than that of PU. As mentioned above, doping GO into PEG would improve the light absorption and shape stability of the composites. At the same time, the thermal stability of the composites could be improved for the interaction of GO with NCO in MDI, and the grafting modification of GO with MDI would improve the thermal conductivity of the composites.

In this work, in order to obtain composite PCMs with the strong light absorption and heat storage capacity at the same time, a novel GO-doped PU (which is prepared by MDI and PEG) composites solid–solid PCMs were prepared by in-situ polymerization (Figure 1). The thermal storage properties, stability, and photothermal properties of PU/GO composites were investigated in a relatively systematic way. The obtained PU/GO composites have high thermal conductivity, superior shape stability, high storage density (150.7 J/g) and outstanding photothermal efficiency (95.3%), which are higher than that (78.7% to 92%) of other PEG-based photo-thermal conversion PCMs. This novel photothermal storage composite PCMs can bring new application prospects for the storage and utilization of solar energy.

2 | EXPERIMENTAL

2.1 | Materials

Flake graphites (NFG, Particle size:300 mesh), potassium permanganate (KMnO_4), and polyethylene glycol (PEG, $M_n = 10\,000$) were all purchased from Sinopharm Chemical Reagent Co. Ltd. Concentrated sulfuric acid (H_2SO_4), hydrochloric acid (HCl), concentrated nitric acid (HNO_3), hydrogen peroxide (H_2O_2), and N, N-Dimethylformamide (DMF) were all obtained from Beijing Chemical Works, Co. Ltd. 4,4'-diphenylmethane diisocyanate (MDI) was from Shanghai Aladdin Bio-Chem Technology Co. Ltd. And 1,4-butanediol (BDO) was supplied from Shanghai Medical Chemical Reagent Co. Inc.

2.2 | Synthesis of PU/GO composites

GO was first prepared from flake graphite according to modified Hummers' method.⁴⁰ The preparation of PU/GO composites were carried out in two steps (Figure 2). Firstly, different masses of MDI, PEG, and DMF were mixed and stirred at 70°C for 1 h to prepare an NCO-terminated prepolymer. Then, GO dispersion was added to the DMF mixture for further reaction for 2 h. In the second step, a stoichiometric amount of the chain extender BDO was slowly put into the solution under continuous stirring and the mixture was maintained at 80°C for 2 h. Finally, the samples were vacuum-dried in a vacuum drying oven at 80°C for 48 h. PU/GO composite PCMs was obtained after vacuum drying. The PCMs with different PEG content (without GO) were fabricated and labeled as PEG, PU1, PU2, and PU3. And the

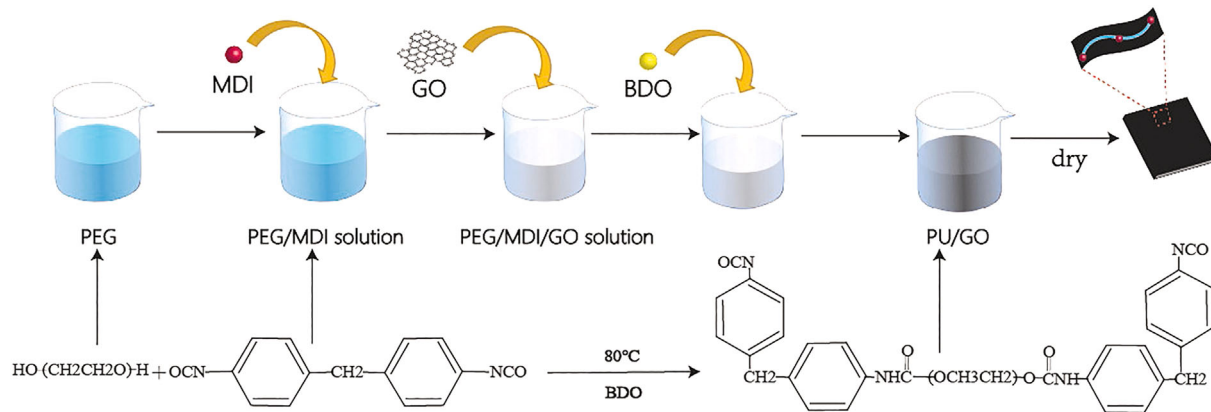


FIGURE 2 Synthetic route of PU/GO

TABLE 1 Compositions of PEG, PU, and PU/GO

Samples	PEG (wt%)	Molar ratio of PEG/MDI	GO (wt%)
PEG	100	/	0
PU1	93	0.48	0
PU2	89	0.31	0
PU3	85	0.2	0
PU1GO1	93	0.48	0.86
PU2GO1	89	0.31	0.86
PU3GO1	85	0.2	0.86
PU2GO2	89	0.31	1.72
PU2GO3	89	0.31	3.33

obtained samples of PU/GO with the same GO, PU content, and different PEG content were labeled as PU1GO1, PU2GO1, and PU3GO1, respectively. In addition, the samples of PU/GO composite PCMs with the same PEG content (same as PU2) and different GO content were labeled as PU2GO1, PU2GO2, and PU2GO3, respectively. The basic recipes were listed in Table 1.

2.3 | Characterization

The morphology and structure of the samples were determined by scanning electron microscope (SEM, Hitachi S-4800) and x-ray diffractometer (XRD, Bruker D8-Advance). The chemical structure of the samples was evaluated by Fourier transform infrared spectroscopy (FT-IR, Brugg Tensor27). Thermal properties were studied by differential scanning calorimeter (DSC, PerkinElmer, at a heating and cooling rate of 10°C/min under 25 mL/min N₂ flow). Heating process and cooling process of samples were conducted on an Agilent-34970A temperature recorder with thermocouple (Agilent Technolo-

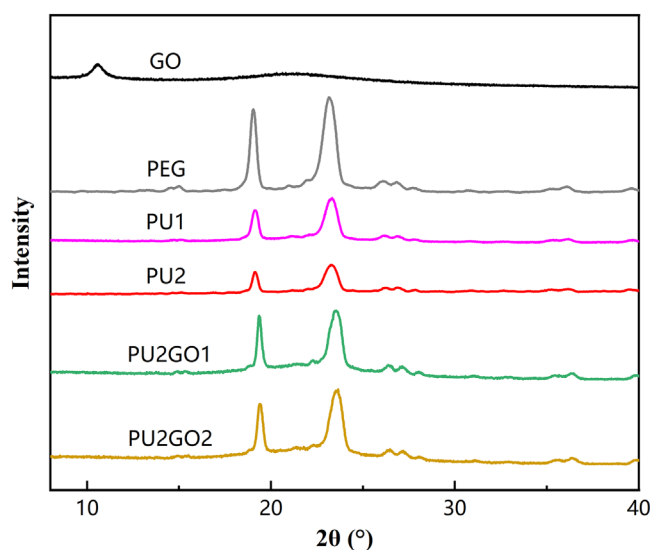


FIGURE 3 XRD patterns of GO, PEG, PU, and PU/GO PCMs

gies [China] Co., Ltd.) and carried out on plate heating. The optical property of the samples was recorded by ultraviolet–visible–near infrared spectrophotometer (UV–Vis–NIR, Cary 5000) at room temperature. In photothermal conversion test, xenon arc lamps (Zolix Co., Ltd.) were applied to simulated sunlight and the NOVAII optical power meter (OPHIR, Israel) recorded light intensity. The photothermal performance evaluation process was performed under simulated solar illumination at a controlled intensity of 130 mW/cm².

3 | RESULTS AND DISCUSSION

3.1 | The chemical and structural analysis of PCMs

Figure 3 shows the XRD patterns of GO, PEG, PU, and PU/GO. The high diffraction peaks of PEG at $2\theta = 19.02^\circ$

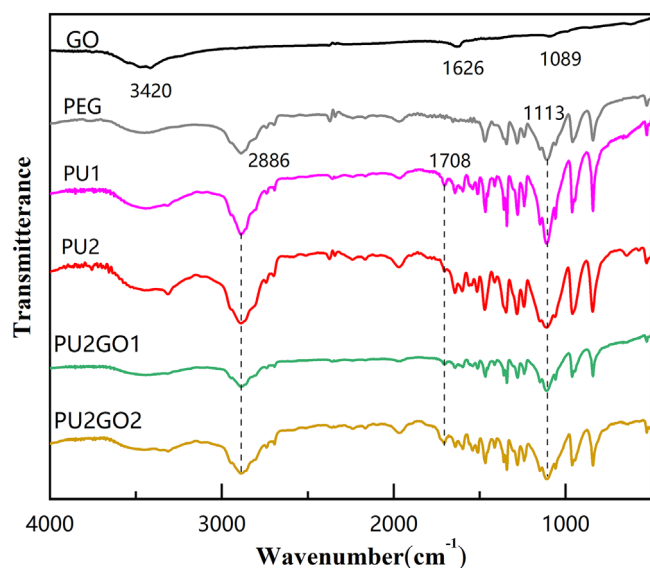


FIGURE 4 FT-IR spectra of GO, PEG, PU, and PU/GO

and 23.15° , which indicate PEG has good crystallinity. After the reaction of PEG with MDI, the apparent decreased intensity of the diffraction peak of PU1 and PU2 indicates that the crystallinity becomes lower. When GO was added into PU2, the intensity of the diffraction peak of PU2GO1 increased obviously. The results indicate that the crystallinity of PU/GO is improved. This is because that some functional groups of GO interacted with -NCO, and the crystallization of PEG could be affected by the crosslinking degrees of PU2, so GO would hinder the interaction between the two sides of the PU2 and reduce the confinement of PEG chain segments in PU2. Hence, the free movement of the PEG chain segments is greatly enhanced, and the crystallinity of PU/GO is improved.³⁴ PU2GO1 exhibits all the characteristic peaks of PU2, and no new diffraction peaks appear. The results indicate the addition of GO is only physical modification and no new substance is produced.

The FT-IR spectra of GO, PEG, PU, and PU/GO composites are shown in Figure 4. In the GO spectrum, the peaks at 3420, 1626, and 1089 cm^{-1} correspond to their O-H, C=O, and C-O tensile vibration. The peaks at 2886 cm^{-1} in the pure PEG spectrum are attributed to the C-H bonds, and the peaks at 1058, 1113, and 1152 cm^{-1} correspond to C-O-C tensile vibration.^{34,41} After the reaction of MDI with PEG, a new absorption peak of -NHCOO- stretching vibration emerges in PU1 and PU2 at 1708 cm^{-1} , indicating that the synthesis of polyurethane is successful.⁴² The strong peaks at 2886 and 1113 cm^{-1} in PU/GO spectra are attributed to the C-H bonds and C-O-C tensile vibration of the soft segment PEG. The spectrums of PU2GO1 and PU2GO2 show the characteristic peaks of PU2 with no new absorption peaks after GO was added. This is mainly

because the content of GO is too small, and even if GO participated in the interaction with MDI, it will not produce other new functional groups.

3.2 | Morphology analysis

Figure 5 shows the morphology of GO, PEG, PU, and PU/GO composites. Figure 5A is the SEM image of GO sheets. It shows that the prepared GO has a folded surface texture and a large lateral size, distributed in layers and stacked on top of each other. Figure 5B–D shows the SEM images of PEG, PU1, and PU2. The image shows that the surface of PU is relatively smoother, and the edges are more obvious than that of PEG. Figure 5E,F shows the SEM images of PU1GO1 and PU2GO1. It was clear that when the MDI content is higher than 10 wt%, the PU surface presented a 2D wrinkled surface owing to the addition of GO sheets. These characteristics may contribute to the shape stability of the resulting composite PCMs. And the surface becomes smoother, and the wrinkles become less with higher content of PEG. When the PEG content is high enough, the morphology of the system returns to a relatively flat irregular strip structure.

3.3 | Thermophysical properties of PCMs

3.3.1 | Thermo-physical properties of PU/GO

Figure 6A,B shows the DSC curves of PEG, PU, and PU/GO composites. The corresponding phase change enthalpies and temperatures of melting and crystallization are summarized in Table 2. When MDI was added into PEG, the phase change enthalpy and the phase change temperature of the system drop dramatically. The results of PU(1-3) and PU(1-3)GO1 show that the melting point (T_m) and crystallization point (T_c) of the composite PCMs were little affected by the PEG content. Figure 6C and Table 2 show that the phase change enthalpies (ΔH) and the degree of crystallinity (X_c) of the PU(1-3) and PU(1-3)GO1 composite PCMs increase with PEG content increases. But the amount of phase change enthalpies (ΔH_m) and the degree of crystallinity (X_c) of the PU(1-3) GO1 composite PCMs is higher than that of PU(1-3) with same PEG content. This is mainly because that GO can enhance the movement of the PEG segment. GO reduced the interaction between the two sides of the PU and the free movement of the PEG chain segments is greatly enhanced. When the PEG content is 93 wt%, the phase change enthalpy (ΔH_m) of the PU1GO1 composite PCMs is as high as 157.2 J/g , which is 32.76 J/g higher than that of PU1 and is close to PEG. This is mainly due to the

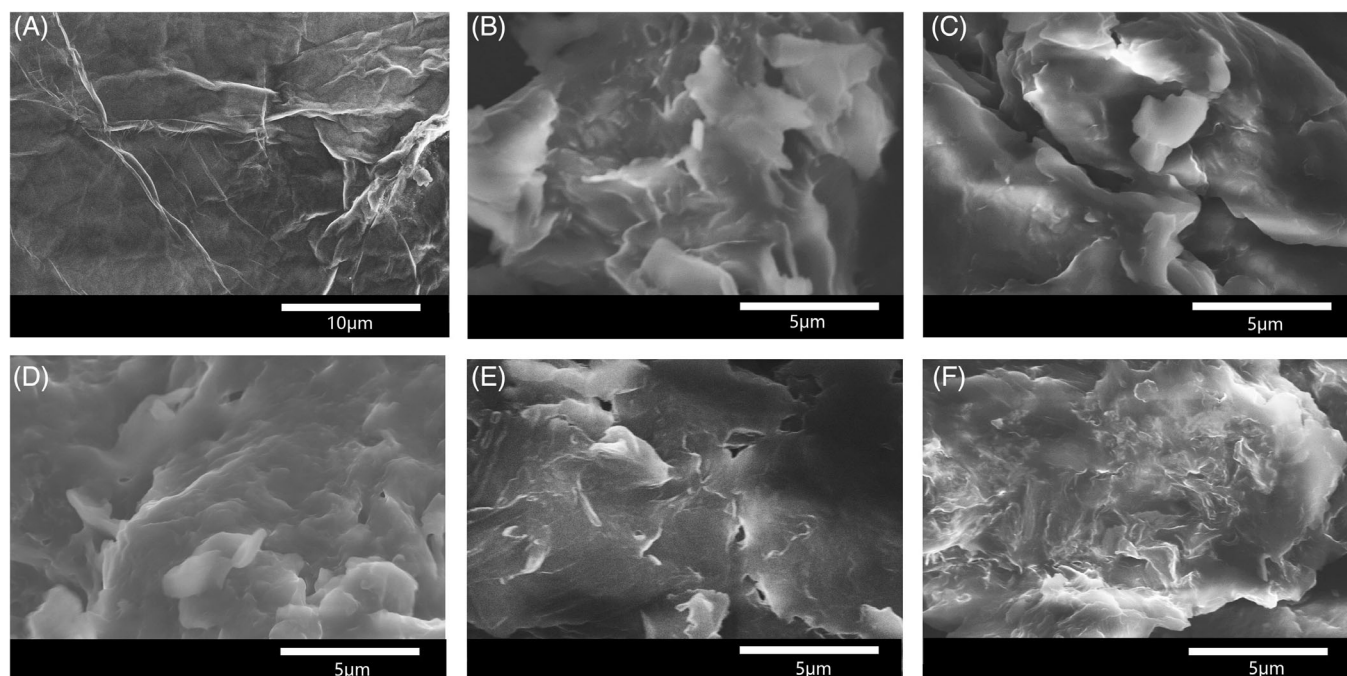


FIGURE 5 SEM images of the surface: (A) GO (B) PEG (C) PU1 (D) PU2 (E) PU1GO1 (F) PU2GO1

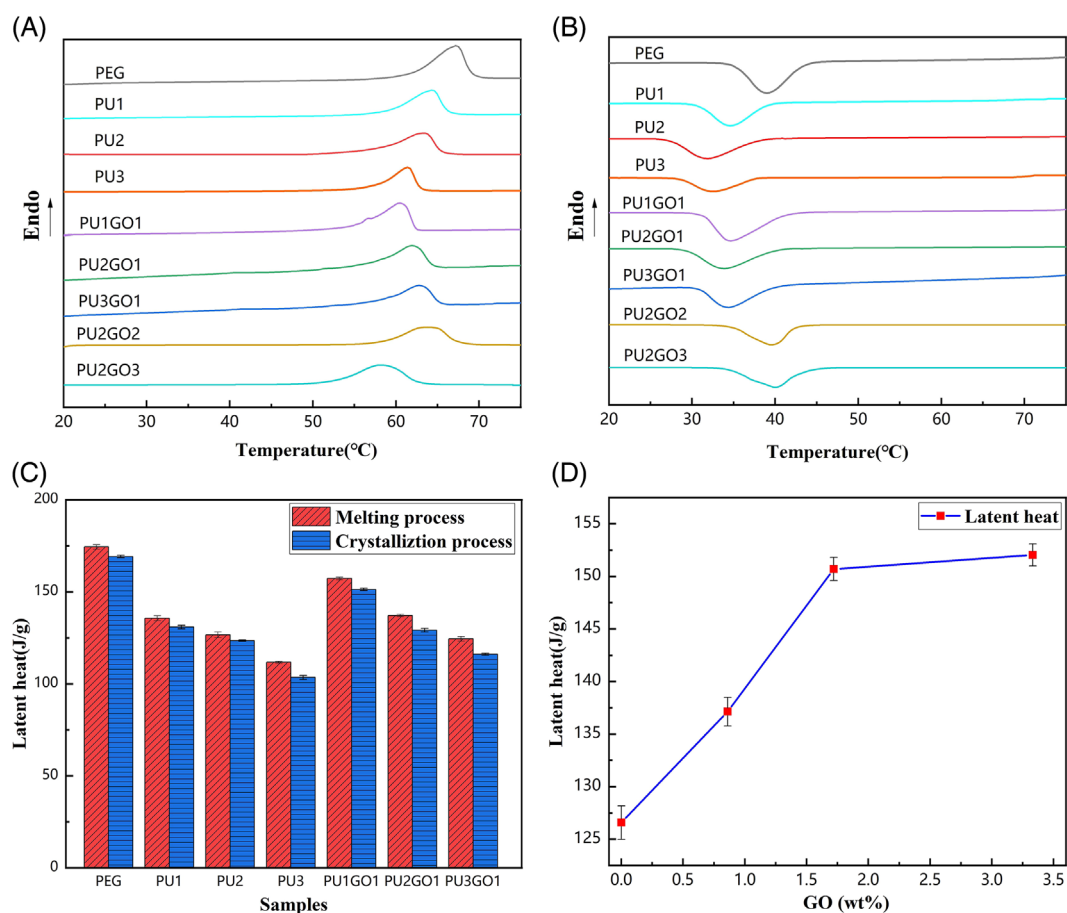


FIGURE 6 DSC curves of PU/GO during (A) heating and (B) cooling (C) Phase change properties of PEG, PU(1-3) and PU(1-3)GO1 composite PCMs (D) Phase change properties of PU2 with different GO content

TABLE 2 DSC data of PEG, PU, and PU/GO

Sample	GO (wt%)	PEG (wt%)	T_m (°C)	T_c (°C)	ΔH_m (J/g)	ΔH_c (J/g)	X_c (%)	ΔT (°C)
PEG	0	100	65.1 ± 0.2	39.3 ± 0.3	174.6 ± 0.8	169.2 ± 0.7	100	25.8
PU1	0	93	64.2 ± 0.2	34.9 ± 0.2	135.8 ± 0.5	130.4 ± 0.6	85.4	29.31
PU2	0	89	63.7 ± 0.3	31.4 ± 0.3	126.5 ± 0.6	123.4 ± 0.5	81.5	32.39
PU3	0	85	61.3 ± 0.3	31.1 ± 0.2	111.7 ± 0.5	103.7 ± 0.7	74.8	30.16
PU1GO1	0.86	93	61.4 ± 0.3	34.0 ± 0.4	157.2 ± 0.7	151.3 ± 0.6	98.9	27.34
PU2GO1	0.86	89	61.9 ± 0.2	33.8 ± 0.5	137.1 ± 0.6	129.1 ± 0.5	88.3	28.12
PU3GO1	0.86	85	62.5 ± 0.4	34.4 ± 0.3	124.4 ± 0.8	116.1 ± 0.7	83.3	28.01
PU2GO2	1.72	89	63.8 ± 0.3	39.5 ± 0.4	150.7 ± 0.9	144.3 ± 0.8	97.1	24.28
PU2GO3	3.33	89	57.9 ± 0.4	40.9 ± 0.5	152.0 ± 1.1	139.8 ± 0.7	97.9	16.91

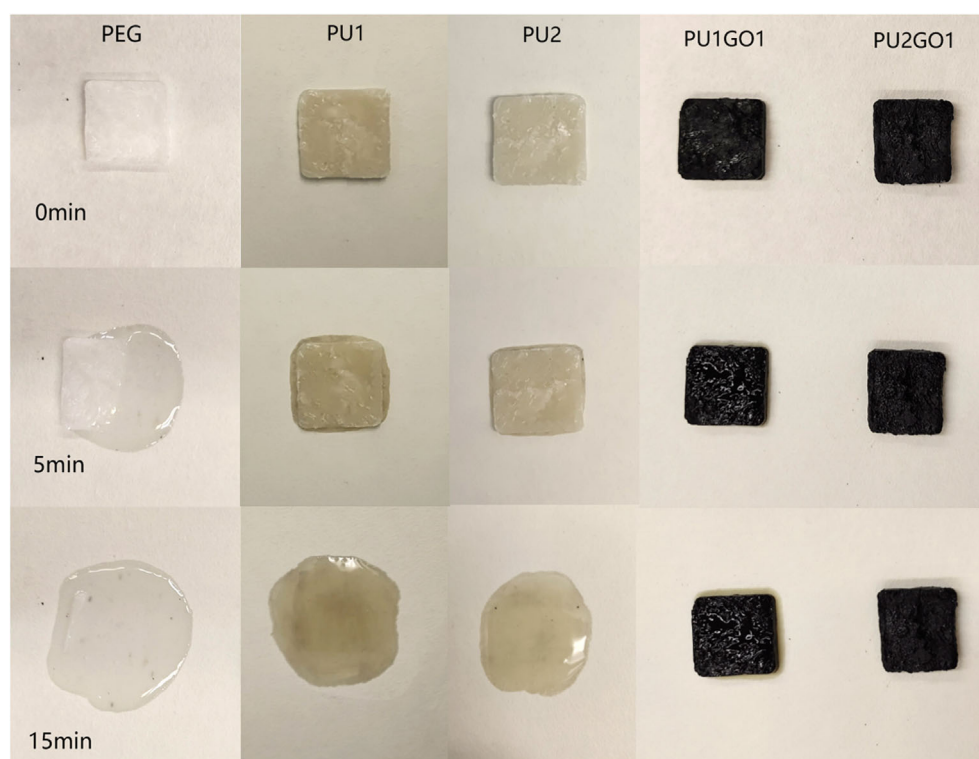


FIGURE 7 Digital images of PEG, PU, and PU/GO at 80°C for 15 min

main contribution of polyol. The increase in soft segment content also enhances the alignment and crystallinity of the molecular chains. The results of PU2GO(1-3) show that, the super-cooling extent (ΔT) of PU/GO can be effectively suppressed by the increase of GO content. This is mainly because that GO forms a continuous heat conduction channel for PEG.⁴³ Figure 6D and Table 2 show that the phase change enthalpies (ΔH_m) and the degree of crystallinity (X_c) of PU2GO(1-3) increase with the increase of GO content but then reach saturation. Because the physical confinement on the soft segment caused by excessive GO is greater than its enhancement. These results indicate that the thermal energy storage capacity of PU/GO composites can be greatly improved

by reasonably modulating the content of GO and soft segments.

3.3.2 | Thermal stability of PU/GO

The thermal stability of PCMs can prevent leakage and encapsulation problems during the phase transition process, so it is critical in practical application. Figure 7 shows a photograph of samples heated on a heating plate at different times. Pure PEG gradually melts and flows when the heating time reaches 5 min. This indicates that when the sample is heated, the soft segments of the crystalline part begin to melt into irregular amorphous

segments and tend to move freely. PU1 starts to melt while PU2 remains its form in 5 min. This is mainly because the hard segments can hold soft segments and limit their free movement in a certain region by the action of chemical bonds and physical restriction. Hence, during the melting process, the system appears as a macroscopic solid state and more hard segments content will lead to less leakage. All PU/GO composites maintain their original shapes in 15 min. This is mainly due to the addition of MDI and GO restricting the fluidity of PEG. PU2GO1 has no leakage while PU1GO1 melts and infiltrates the filter paper slightly. The result indicates that soft segment with 89 wt% content has better form-stability, which can be demonstrated by the SEM image (Figure 5D,E). Therefore, we chose PU2 as the main sample of the solid-solid PCMs for further research.

TGA curves of samples are shown in Figure 8 and the characteristic weight loss temperatures are listed in Table 3. For composite PCMs, no significant weight loss was observed in the temperature range of 60–300°C, indicating that composite PCMs are stable at phase change temperature (61–64°C). The composites have a very slight decomposition between 300–350°C corresponding to the

rupture of the urethane in the PU. With addition of GO, its initial temperature at 5% weight loss increases from 343°C to 379°C, indicating that PU/GO composites have better thermal stability than pure PU. This may be due to the reaction of -NCO with -OH or -COOH of GO.⁴⁴ The results of maximum weight loss of PU2GO(1-3) show that, the maximum mass loss will gradually decrease with the increase of GO content. This is mainly due to the high interfacial adhesion between NCO-GO and PU, which suppresses the decomposition of the PU/GO composites.³⁸

3.3.3 | Thermal conductivity of PU/GO

Figure 9A displays the thermal conductivities of PEG, PU, and the PU/GO composite PCMs. PU(1-3)GO1 composite PCMs with 0.86 wt% GO exhibits much higher thermal conductivity than that of PU(1-3) with the same PEG content, respectively. The thermal conductivity of the PU(1-3) and PU(1-3)GO1 composite PCMs increases with the decrease in the content of PEG in PU. This is mainly because the addition of much PEG leads to an increase in the voids of the composite material. It is the key point to hinder the formation of thermal conduction network in PU/GO composite PCMs, which results in a decrease in thermal conductivity. Figure 9B shows that the thermal conductivity of the PU2GO(1-3) composite PCMs gradually increased with the increase of GO content. The thermal conductivity of PU2GO2 can reach as high as $0.972 \text{ Wm}^{-1} \text{ K}^{-1}$, which shows relative enhancement of 188.4% higher than that of PEG. The -NCO in MDI can be covalently grafted with GO and then greatly help to reduce the interface thermal resistance, which greatly improve the thermal conductivity of PU/GO composite PCMs.^{45,46} However, the thermal conductivity of PU2GO3 is lower than that of PU2GO2. This is mainly because that the addition of excessive content of GO will cause the agglomeration of fillers in PU/GO, resulting in the increased interfacial thermal resistance and blocked phonon transmission, which has an impact on thermal energy transfer.^{47,48}

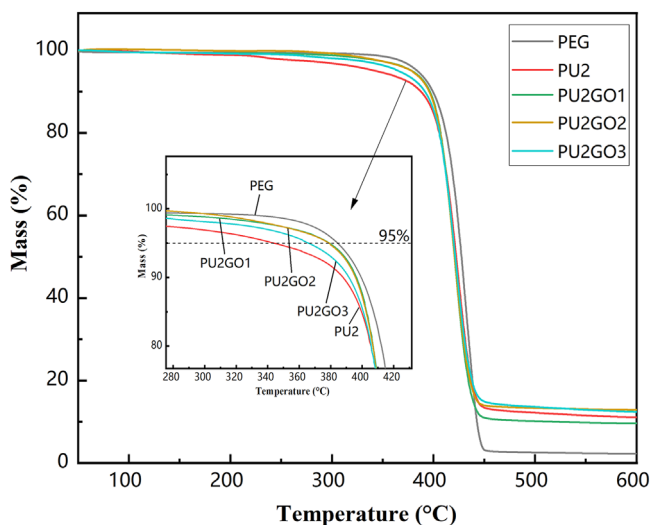


FIGURE 8 TGA curves of PEG, PU, and PU/GO

TABLE 3 TGA data of PEG, PU, and PU/GO in nitrogen

Sample	Temperature at 5% weight loss (°C)	Temperature at 10% weight loss (°C)	Temperature at maximum weight loss rate (°C)	Maximum weight loss(%)
PEG	384	399	451	97.0
PU2	343	387	447	86.2
PU2GO1	379	396	446	88.5
PU2GO2	379	395	446	85.8
PU2GO3	368	391	450	85.2

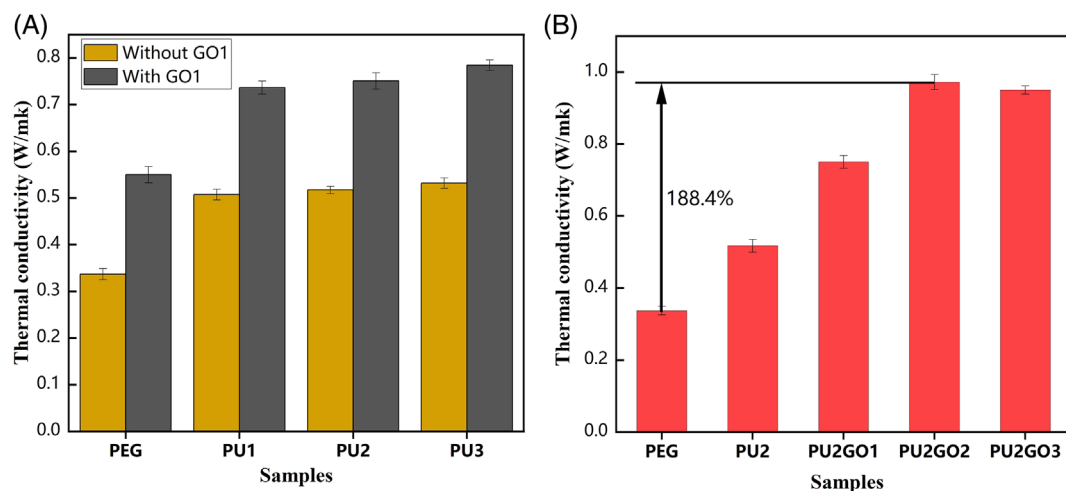


FIGURE 9 (A) Comparison of thermal conductivity of PCMs without GO and with GO (B) Thermal conductivities of PEG, PU, and PU/GO

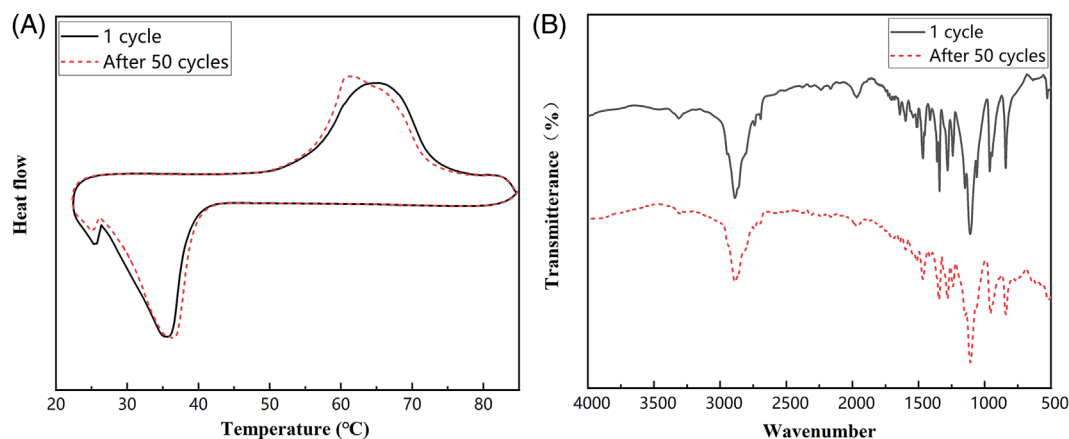


FIGURE 10 (A) Heating and cooling DSC curves of PU2GO2 before and after 50 thermal cycles (B) FTIR spectrum of PU2GO2 before and after 50 thermal cycles

3.3.4 | Thermal reliability of PU/GO

Figure 10 shows the DSC curves and FTIR spectra of PU2GO2 before and after 50 thermal cycles. Small fluctuations in the heat of melt of PU/GO PCMs were observed after 50 phase transition cycles. The phase change enthalpy was reduced by 5.2 J/g, and the enthalpy retention was 96.2%. And no obvious changes in the position of the characteristic peaks from FTIR spectra can be observed. These results show PU/GO composites have an outstanding thermal stability and stable chemical structure.

3.3.5 | Thermoregulation performance of PU/GO

Figure 11A shows the temperature changes of PEG, PU, and PU/GO composite PCMs with time during the cooling process. PEG, PU, and PU/GO were heated to 60°C

and then removed the heat source. The release times required are 420 s, 1130s, and 2015s from 60°C to 30°C for PEG, PU, and PU/GO. Figure 11B shows that it took 3.8 times longer for PU/GO to reach 30°C than PEG. The results show that PU/GO composites have great thermoregulation performance and can mitigate the reduction of temperature.

3.4 | Photothermal energy conversion

The PU/GO composites can achieve photothermal energy conversion and thermal energy storage owing to the light-absorbing property of GO and the latent energy storage of PEG. As shown in the UV-vis absorption spectra (Figure 12A), the addition of low content of GO would greatly improve the light absorption capacity of PU2GO2 in a wide wavelength range, which is important for effective light capture and photothermal conversion.

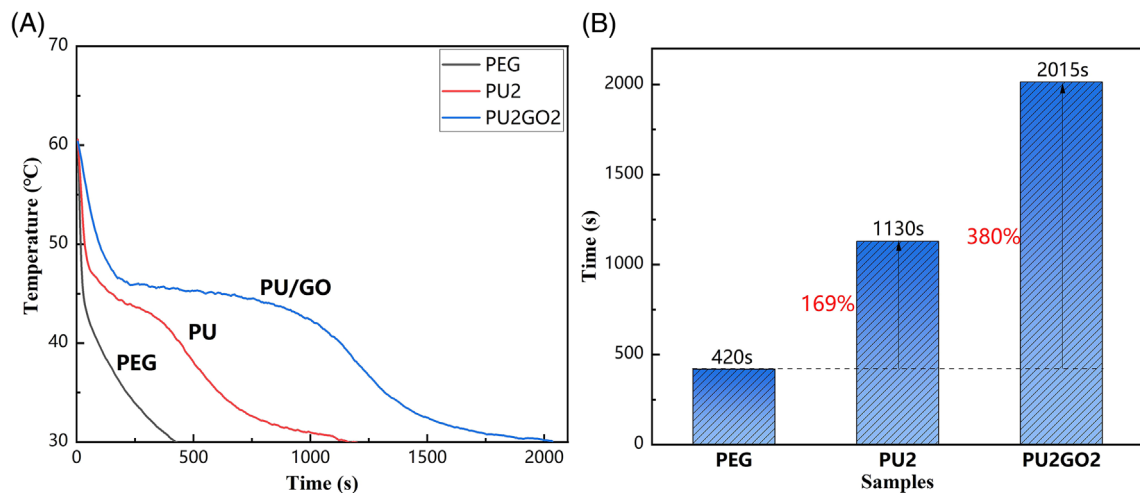


FIGURE 11 (A) Thermal energy release curves for PEG, PU2, and PU2GO2 (B) Thermal energy release time of PEG, PU2, and PU2GO2

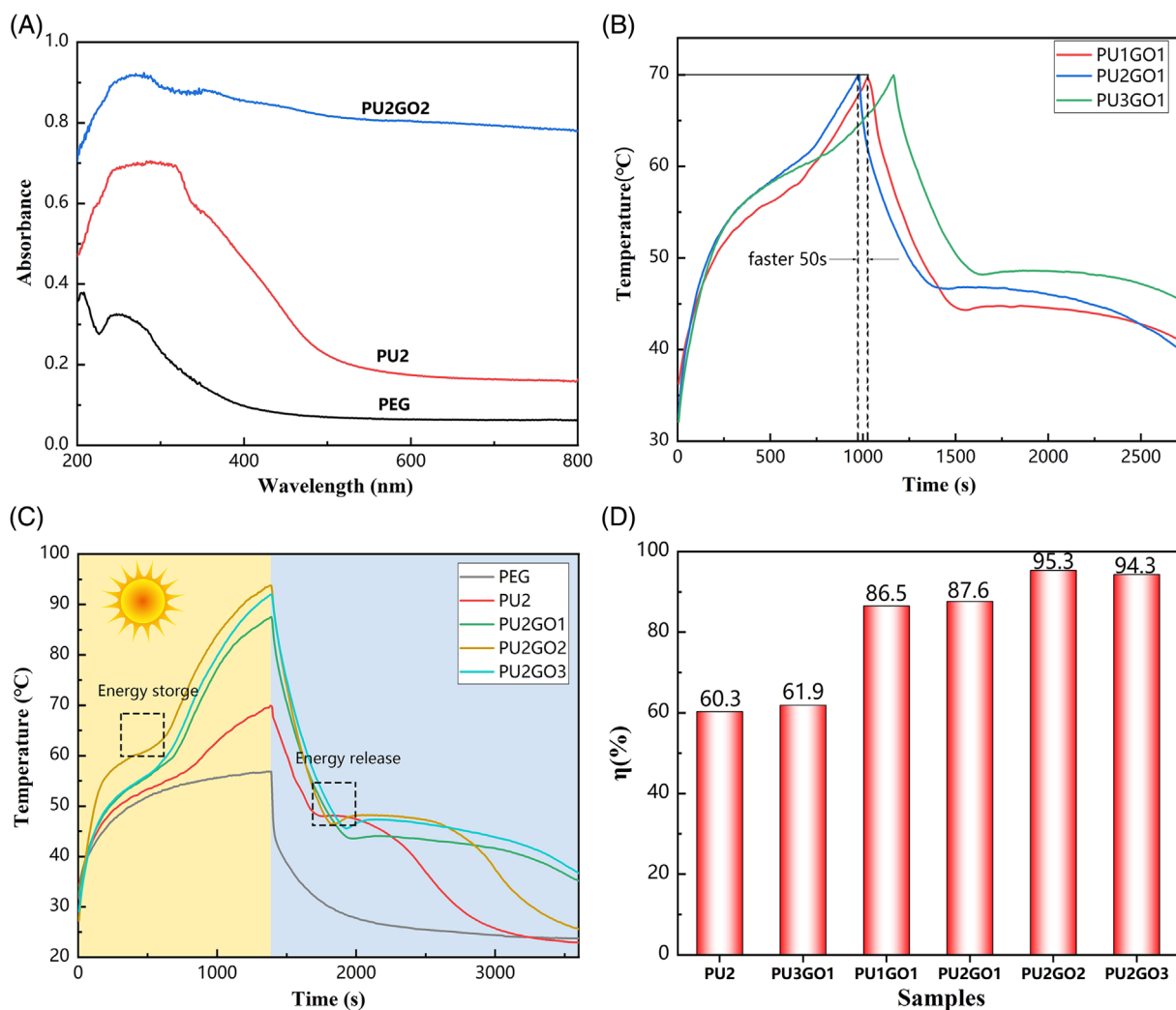


FIGURE 12 (A) UV-VIS-NIR absorption spectrum of PEG, PU, and PU2GO2 (B,C) Photothermal conversion curves (D) Photothermal conversion efficiency of PU and PU2GO2

TABLE 4 Comparison of thermal performance and photothermal performance of different PCMs

Sample	Thermal conductivity (Wm ⁻¹ K ⁻¹)	Phase change enthalpy (J/ g)	Photothermal energy conversion efficiency (%)	Reference
PEG/Ag-GNS	0.414	166.1	92	26
PEG/HDIB/RGO	/	107.2	78.7	34
PEG/GO	/	84.27	87.3	36
PEG/GO/CF	0.94	119.01	84.2	49
PEG/BN/GNP	1.33	123.8	72.7	50
PEG/RGO-CoO	0.523	131.94	/	25
PEG/SiO ₂ -dye	/	167	88	51
PEG/MDI/RGO	0.696	120.8	81.8	52
PEG/MDI/GO	0.972	150.7	95.3	This work

Under simulated solar radiation, temperature of the samples increased after solar radiation. When the temperature reached their melting point, a phase change platform of PU/GO appeared, corresponding to the solid-liquid phase transition and the storage of solar energy. After turning off the light, the temperature of composites PCMs dropped sharply. When the temperature reached the crystallization point, another platform emerged. It shows that PU/GO starts to solidify and release energy. It can be seen from Figure 12B that the heating rate increases with the increase of PEG content in the PU(1-3)GO1 composites. However, the temperature of PU2GO1 rises from 30°C to 70°C in 975 s, which is 50s faster than PU1GO1 and 190 s faster than PU3GO1. This indicates that the best photothermal performance of PU(1-3)GO1 is achieved when the content of PEG is 89 wt%. As mentioned above, PU2GO1 with 89 wt% of PEG content is also the most suitable ratio for solid-solid phase transition performance. This is due to the thermal conductivity influence on heat transfer performance, and the heat transfer performance decreases with the increase of PEG content.⁴⁸ Furthermore, the photothermal conversion of PU2/GO composites with different contents of GO (0.86, 1.72, and 3.33 wt%) is shown in Figure 12C. It can be found that the temperatures of the PU2/GO composites increase faster with increasing GO content. And the temperature plateau becomes shorter with increasing GO content. However, the heating rate of PU2GO2 is faster than that of PU2GO3, which is consistent with the previous results of thermal conductivity and phase change enthalpy. This may be due to uneven distribution and aggregation caused by excessive GO filling in PU2GO3. The efficiency of photothermal conversion can be calculated by the following equation:

$$\eta = \frac{m\Delta H}{PA(t_t - t_f)} \quad (1)$$

where m is the mass of the sample, and ΔH is the enthalpy of phase transition, P is the intensity of light, A is the surface area of samples, t_t and t_f are the start time and end time of phase transition, respectively. As is shown in Figure 12D, when PEG content is 89 wt% and GO content is 1.72 wt% (PU2GO2), η is as high as 95.3%. It is 35% higher than that of PU2. The excellent heat transfer and energy storage capability are obtained by the rational modulation of the hard and soft segments. Some of the reported thermal conductivities and solar photothermal conversion efficiencies of PEG-based composite PCMs are listed in Table 4. In comparison with other PEG-based PCMs, the photo-thermal conversion efficiency (95.3%) of this study is found to be better. At the same time, it also has a high phase change enthalpy and thermal conductivity. These results confirm that PU/GO can efficiently absorb and convert sunlight into thermal energy.

4 | CONCLUSION

In this study, PU/GO composites with high molecule weight PEG as soft segment were synthesized by two steps. With the addition of MDI and GO, the thermal conductivity and photothermal efficiency of PU/GO composites were significantly increased, while the phase change enthalpy also improved due to the increase of soft segment content. Thus, the energy storage capability, stability, thermal reliability, and photothermal conversion efficiency of PU/GO composites solid-solid PCMs were all enhanced. The results show that the thermal conductivity of PU/GO is 53.4% to 188.4% higher than that of PEG. It took 3.8 times longer for PU/GO from 60°C to 30°C than PEG, indicating PU/GO has great thermoregulation performance. And the introduction of low content of GO also can help the PU/GO composite PCMs realize

strong light absorption ability, which helps to achieve efficient photothermal conversion. The PU/GO composite doped with 89 wt% PEG and 1.72 wt% GO shows the highest heating rate and photothermal conversion efficiency is 95.3%. The 2D structure of NCO-GO can provide a new way for the solar energy storage and conversation.

ACKNOWLEDGEMENTS

The research was funded by the National Natural Science Foundation of China (No. 52176081, 51876111), by Shanghai Rising-Star Program (No. 21QA1403300), by Shanghai Local Capacity Building Program (No. 22010500700, 22010500500), by Natural Science Foundation of Shanghai (No. 21ZR1424500), the “Shu Guang” Project of Shanghai Municipal Education Commission and Shanghai Education Development Foundation (Grant No. 18SG54).

ORCID

Jiawei Wang  <https://orcid.org/0000-0002-9715-9944>

Zihua Wu  <https://orcid.org/0000-0002-0369-3151>

REFERENCES

- Sharshir SW, Elsheikh AH, Edreis EMA, et al. Improving the solar still performance by using thermal energy storage materials: a review of recent developments. *Desalination Water Treat.* 2019;165:1-15.
- Dalvi VH, Panse SV, Joshi JB. Solar thermal technologies as a bridge from fossil fuels to renewables. *Nat Clim Change.* 2015; 5(11):1007-1013.
- Liu X, Iocozzia J, Wang Y, et al. Noble metal-metal oxide nanohybrids with tailored nanostructures for efficient solar energy conversion, photocatalysis and environmental remediation. *Energ Environ Sci.* 2017;10(2):402-434.
- Zheng X, Zhang L. Photonic nanostructures for solar energy conversion. *Energ Environ Sci.* 2016;9(8):2511-2532.
- Khan Z, Khan Z, Ghafoor A. A review of performance enhancement of PCM based latent heat storage system within the context of materials, thermal stability and compatibility. *Energ Conver Manage.* 2016;115:132-158.
- Rathod MK, Banerjee J. Thermal stability of phase change materials used in latent heat energy storage systems: a review. *Renew Sustain Energy Rev.* 2013;18:246-258.
- Yang J, Qi G, Liu Y, et al. Hybrid graphene aerogels/phase change material composites: thermal conductivity, shape-stabilization and light-to-thermal energy storage. *Carbon.* 2016; 100:693-702.
- Wang XL, Li B, Qu ZG, Zhang JF, Jin ZG. Effects of graphite microstructure evolution on the anisotropic thermal conductivity of expanded graphite/paraffin phase change materials and their thermal energy storage performance. *Int J Heat Mass Trans.* 2020;155:119853.
- Alkan C, Günther E, Hiebler S, Himpel M. Complexing blends of polyacrylic acid-polyethylene glycol and poly(ethylene-co-acrylic acid)-polyethylene glycol as shape stabilized phase change materials. *Energ Conver Manage.* 2012;64:364-370.
- Onder E, Sarier N, Ukuser G, Ozturk M, Arat R. Ultrasound assisted solvent free intercalation of montmorillonite with PEG1000: a new type of organoclay with improved thermal properties. *Thermochim Acta.* 2013;566:24-35.
- Tanwar S, Kaur R. Development and investigation of microencapsulated caprylic acid-based phase change materials for thermal energy storage applications. *Int J Energy Res.* 2021;45(12): 17302-17314.
- Wang J, Han W, Ge C, Guan H, Yang H, Zhang X. Form-stable oxalic acid dihydrate/glycolic acid-based composite PCMs for thermal energy storage. *Renew Energy.* 2019;136:657-663.
- Sarı A, Biçer A, Alkan C. Thermal energy storage characteristics of poly(styrene-co-maleic anhydride)-graft-PEG as polymeric solid-solid phase change materials. *Sol Energy Mater Sol Cells.* 2017;161:219-225.
- Xi P, Gu X, Cheng B, Wang Y. Preparation and characterization of a novel polymeric based solid-solid phase change heat storage material. *Energ Conver Manage.* 2009;50(6):1522-1528.
- Zhou Z, Xu W, Fan J, Ren F, Xu C. Synthesis and characterization of carboxyl group-containing acrylic resin for powder coatings. *Prog Org Coat.* 2008;62(2):179-182.
- Pomianowski M, Heiselberg P, Zhang Y. Review of thermal energy storage technologies based on PCM application in buildings. *Energ Buildings.* 2013;67:56-69.
- Jia W, Wang C, Wang T, Cai Z, Chen K. Preparation and performances of palmitic acid/diatomite form-stable composite phase change materials. *Int J Energy Res.* 2020;44(6):4298-4308.
- Stropanik R, Koželj R, Zavr E, Stritih U. Improved thermal energy storage for nearly zero energy buildings with PCM integration. *Sol Energy.* 2019;190:420-426.
- Chen L, Zou R, Xia W, et al. Electro- and photodriven phase change composites based on wax-infiltrated carbon nanotube sponges. *ACS Nano.* 2012;6(12):10884-10892.
- Tang Y, Jia Y, Alva G, Huang X, Fang G. Synthesis, characterization and properties of palmitic acid/high density polyethylene/graphene nanoplatelets composites as form-stable phase change materials. *Sol Energy Mater Sol Cells.* 2016;155:421-429.
- Sarı A, Bicer A, Al-Ahmed A, Al-Sulaiman FA, Zahir MH, Mohamed SA. Silica fume/capric acid-palmitic acid composite phase change material doped with CNTs for thermal energy storage. *Sol Energy Mater Sol Cells.* 2018;179:353-361.
- Noureen L, Xie Z, Gao Y, et al. Multifunctional Ag₃PO₄-rGO-coated textiles for clean water production by solar-driven evaporation, Photocatalysis, and Disinfection. *ACS Appl Mater Interfaces.* 2020;12(5):6343-6350.
- Tang B, Wei H, Zhao D, Zhang S. Cells, light-heat conversion and thermal conductivity enhancement of PEG/SiO₂ composite PCM by in situ Ti₄O₇ doping. *Sol Energy Mater Sol Cells.* 2017; 161:183-189.
- Wang W, Umair MM, Qiu J, et al. Electromagnetic and solar energy conversion and storage based on Fe₃O₄-functionalised graphene/phase change material nanocomposites. *Energ Conver Manage.* 2019;196:1299-1305.
- Li Y, Li Y, Huang X, et al. Graphene-CoO/PEG composite phase change materials with enhanced solar-to-thermal energy conversion and storage capacity. *Compos Sci Technol.* 2020;195: 108197.
- Zhang Y, Wang J, Qiu J, et al. Ag-graphene/PEG composite phase change materials for enhancing solar-thermal energy conversion and storage capacity. *Appl Energy.* 2019;237:83-90.

27. Balandin AA, Ghosh S, Bao W, et al. Superior thermal conductivity of single-layer graphene. *Nano Lett.* 2008;8(3):902-907.
28. Zhao J, Luo W, Kim J-K, Yang J. Graphene oxide aerogel beads filled with phase change material for latent heat storage and release. *ACS Appl Energy Mater.* 2019;2(5):3657-3664.
29. Wang W, Tang B, Ju B, Gao Z, Xiu J, Zhang S. Fe₃O₄-functionalized graphene nanosheet embedded phase change material composites: efficient magnetic- and sunlight-driven energy conversion and storage. *J Mater Chem A.* 2017;5(3):958-968.
30. Yu W, Sisi L, Haiyan Y, Jie L. Progress in the functional modification of graphene/graphene oxide: a review. *RSC Adv.* 2020;10(26):15328-15345.
31. You M, Zhang X, Wang X, Zhang L, Wen W. Effects of type and contents of microencapsulated n-alkanes on properties of soft polyurethane foams. *Thermochim Acta.* 2010;500(1-2):69-75.
32. Bernard C, Goodwin DG Jr, Gu X, et al. Graphene oxide/waterborne polyurethane nanocoatings: effects of graphene oxide content on performance properties. *J Coat Technol Res.* 2020;17(1):255-269.
33. Cai D, Jin J, Yusoh K, Rafiq R, Song M. High performance polyurethane/functionalized graphene nanocomposites with improved mechanical and thermal properties. *Compos Sci Technol.* 2012;72(6):702-707.
34. Zhou Y, Liu X, Sheng D, et al. Polyurethane-based solid-solid phase change materials with in situ reduced graphene oxide for light-thermal energy conversion and storage. *Chem Eng J.* 2018;338:117-125.
35. Tang L, Yang J, Bao R, et al. Polyethylene glycol/graphene oxide aerogel shape-stabilized phase change materials for photo-to-thermal energy conversion and storage via tuning the oxidation degree of graphene oxide. *Energ Conver Manage.* 2017;146:253-264.
36. Li M, Wang C. Preparation and characterization of GO/PEG photo-thermal conversion form-stable composite phase change materials. *Renew Energy.* 2019;141:1005-1012.
37. Bao L, Liu Z. Near-infrared absorption photothermal conversion polyurethane film for energy storage. *J. Polym. Res.* 2021;28(21):1-10.
38. Zheng F, Jiang P, Hu L, Bao Y, Xia J. Functionalization of graphene oxide with different diisocyanates and their use as a reinforcement in waterborne polyurethane composites. *J Macromol Sci A.* 2019;56(12):1071-1081.
39. Kong D, Xin C, Zhao L, Zhang Z, Wang P, Li Z. Modification of polyurethane by covalently functionalized graphene oxide-with 4,4'-diphenylmethane diisocyanate. *China Synthetic Rubber Ind.* 2021;44(3):230.
40. Marcano DC, Kosynkin DV, Berlin JM, et al. Improved synthesis of graphene oxide. *ACS Nano.* 2010;4(8):4806-4814.
41. Li Y, Sun K, Kou Y, et al. One-step synthesis of graphene-based composite phase change materials with high solar-thermal conversion efficiency. *Chem Eng J.* 2022;429:132439.
42. Alkan C, Günther E, Hiebler S, Ensari ÖF, Kahraman D. Polyurethanes as solid-solid phase change materials for thermal energy storage. *Sol Energy.* 2012;86(6):1761-1769.
43. Wang C, Feng L, Li W, Zheng J, Tian W, Li X. Shape-stabilized phase change materials based on polyethylene glycol/porous carbon composite: the influence of the pore structure of the carbon materials. *Sol Energy Mater Sol Cells.* 2012;105:21-26.
44. Saha U, Jaiswal R, Singh JP, Goswami TH. Diisocyanate modified graphene oxide network structure: steric effect of diisocyanates on bimolecular cross-linking degree. *J Nanopart Res.* 2014;16(2404):1-21.
45. Cao H, Tan Z, Fan G, et al. Wide and fine alignment control and interface modification for high-performance thermally conductive graphite/copper composite. *Compos Part B Eng.* 2020;191:107965.
46. Zhao Y, Jin L, Zou B, et al. Expanded graphite – paraffin composite phase change materials: effect of particle size on the composite structure and properties. *Appl Therm Eng.* 2020;171:115015.
47. Li C, Zheng Y, Wang Z. In-situ polymerization and properties of reduced graphene oxide/thermoplastic polyurethane composites. *Chem Eng Equipment.* 2020;5:3-8.
48. Yuan J, Qian X, Meng Z, Yang B, Liu ZQ. Highly thermally conducting polymer-based films with magnetic field-assisted vertically aligned hexagonal boron nitride for flexible electronic encapsulation. *ACS Appl Mater Interfaces.* 2019;11(19):17915-17924.
49. Zhang H, Wang L, Xi S, Xie H, Yu W. 3D porous copper foam-based shape-stabilized composite phase change materials for high photothermal conversion, thermal conductivity and storage. *Renew Energy.* 2021;175:307-317.
50. Yang J, Tang L-S, Bao R-Y, et al. Largely enhanced thermal conductivity of poly (ethylene glycol)/boron nitride composite phase change materials for solar-thermal-electric energy conversion and storage with very low content of graphene nanoplatelets. *Chem Eng J.* 2017;315:481-490.
51. Yan D, Tang B, Zhang S. Preparation and performances of sunlight-induced phase change PEG/SiO₂-dye composite for solar energy conversion and storage. *Sol Energy Mater Sol Cells.* 2020;215:110657.
52. Mu B, Li M. Fabrication and characterization of polyurethane-grafted reduced graphene oxide as solid-solid phase change materials for solar energy conversion and storage. *Sol Energy.* 2019;188:230-238.

How to cite this article: Wang J, Wu Z, Xie H, et al. Graphene oxide/polyurethane-based composite solid–solid phase change materials with enhanced energy storage capacity and photothermal performance. *Int J Energy Res.* 2022; 46(15):22744-22756. doi:10.1002/er.8576